AWS Bastion Host Lab using GUI + MobaXterm

Overview

This guide walks you from an existing VPC to Bastion Host and Private EC2 access using AWS Console GUI and MobaXterm.

Prerequisites

AWS account, existing VPC, Windows laptop, MobaXterm installed, downloaded .pem key pair.

Step 1: Verify Existing VPC

AWS Console > VPC > Your VPCs. Note VPC ID, CIDR, Region.

Step 2: Create Subnets

Create Public Subnet (10.0.1.0/24) and Private Subnet (10.0.2.0/24) in same VPC.

Step 3: Internet Gateway

Create or verify IGW attached to VPC.

Step 4: Public Route Table

Create route table, add 0.0.0.0/0 to IGW, associate Public Subnet.

Step 5: Security Groups

BastionSG: SSH from My IP. PrivateSG: SSH from BastionSG only.

Step 6: Key Pair

EC2 > Key Pairs > Create key pair bastion-key.pem.

Step 7: Launch Bastion Host

Launch EC2 in Public Subnet with public IP enabled and BastionSG.

Step 8: Launch Private Server

Launch EC2 in Private Subnet with public IP disabled and PrivateSG.

Step 9: Connect via MobaXterm

Session > SSH. Host=Bastion Public IP, user=ec2-user, private key=bastion-key.pem.

Step 10: Upload Key to Bastion

Use left file panel drag-drop bastion-key.pem to /home/ec2-user/ then chmod 400 bastion-key.pem.

Step 11: Connect to Private Server

Inside bastion: ssh -i bastion-key.pem ec2-user@PRIVATE_IP

Step 12: Direct Gateway Session

Create SSH session to PRIVATE_IP and enable SSH gateway using Bastion Public IP.

Troubleshooting

Check usernames, SG rules, key permissions, instance running state.

Best Practices

Restrict Bastion SG to your IP, prefer gateway mode, stop bastion when unused, rotate keys.

Final Architecture

Laptop -> Bastion Host -> Private Server

Useful Commands

- chmod 400 bastion-key.pem
- ssh -i bastion-key.pem ec2-user@BASTION_PUBLIC_IP
- ssh -i bastion-key.pem ec2-user@PRIVATE_IP